



 **cardiffnlp/twitter-roberta-base-sentiment-latest** 

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 Text Classification

 Transformers

 PyTorch

 TensorFlow

 tweet_eval

 English

roberta

 arxiv:2202.03829

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Deploy 

Use this model 

 **Model card**

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 Text Classification

Examples 



Generate

 View Code

 Maximize

 Model tree for cardiffnlp/twitter-roberta-base-sentiment-latest

 Adapters	11 models
 Finetunes	225 models
 Quantizations	3 models

 **Dataset used to train cardiffnlp/twitter-roberta-base-sentiment-latest**

 **cardiffnlp/tweet_eval** Viewer • Updated Jan 4, 2024 •  201k •  19.1k •  139

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Paper for cardiffnlp/twitter-roberta-base-sentiment-latest

TimeLMs: Diachronic Language Models from Twitter

 Paper • 2202.03829 • Published Feb 8, 2022 •  1

Twitter-roBERTa-base for Sentiment Analysis - UPDATED (2022)

This is a RoBERTa-base model trained on ~124M tweets from January 2018 to December 2021, and finetuned for sentiment analysis with the TweetEval benchmark. The original Twitter-based RoBERTa model can be found [here](#) and the original reference paper is [TweetEval](#). This model is suitable for English.

- Reference Paper: [TimeLMs paper](#).
- Git Repo: [TimeLMs official repository](#).

Labels: 0 -> Negative; 1 -> Neutral; 2 -> Positive

This sentiment analysis model has been integrated into [TweetNLP](#). You can access the demo [here](#).

🔗 Example Pipeline

```
from transformers import pipeline
sentiment_task = pipeline("sentiment-analysis", model=model_path, tokenizer=m
sentiment_task("Covid cases are increasing fast!")
```



```
[{'label': 'Negative', 'score': 0.7236}]
```

🔗 Full classification example

```
from transformers import AutoModelForSequenceClassification
from transformers import TFAutoModelForSequenceClassification
from transformers import AutoTokenizer, AutoConfig
import numpy as np
from scipy.special import softmax
# Preprocess text (username and link placeholders)
def preprocess(text):
    new_text = []
    for t in text.split(" "):
        t = '@user' if t.startswith('@') and len(t) > 1 else t
        t = 'http' if t.startswith('http') else t
        new_text.append(t)
    return " ".join(new_text)
MODEL = f"cardiffnlp/twitter-roberta-base-sentiment-latest"
tokenizer = AutoTokenizer.from_pretrained(MODEL)
config = AutoConfig.from_pretrained(MODEL)
# PT
model = AutoModelForSequenceClassification.from_pretrained(MODEL)
```

```

#model.save_pretrained(MODEL)
text = "Covid cases are increasing fast!"
text = preprocess(text)
encoded_input = tokenizer(text, return_tensors='pt')
output = model(**encoded_input)
scores = output[0][0].detach().numpy()
scores = softmax(scores)
# # TF
# model = TFAutoModelForSequenceClassification.from_pretrained(MODEL)
# model.save_pretrained(MODEL)
# text = "Covid cases are increasing fast!"
# encoded_input = tokenizer(text, return_tensors='tf')
# output = model(encoded_input)
# scores = output[0][0].numpy()
# scores = softmax(scores)
# Print labels and scores
ranking = np.argsort(scores)
ranking = ranking[::-1]
for i in range(scores.shape[0]):
    l = config.id2label[ranking[i]]
    s = scores[ranking[i]]
    print(f"{i+1}) {l} {np.round(float(s), 4)}")

```

Output:

- 1) Negative 0.7236
- 2) Neutral 0.2287
- 3) Positive 0.0477

References

```

@inproceedings{camacho-collados-etal-2022-tweetnlp,
  title = "{T}weet{NLP}: Cutting-Edge Natural Language Processing for Social Media Sentiment Analysis",
  author = "Camacho-collados, Jose and
    Rezaee, Kiamehr and
    Riahi, Talayeh and

```

```

    Ushio, Asahi and
    Loureiro, Daniel and
    Antypas, Dimosthenis and
    Boisson, Joanne and
    Espinosa Anke, Luis and
    Liu, Fangyu and
    Mart{\'\i}nez C{\'a}mara, Eugenio" and others,
booktitle = "Proceedings of the 2022 Conference on Empirical Methods in N
month = dec,
year = "2022",
address = "Abu Dhabi, UAE",
publisher = "Association for Computational Linguistics",
url = "https://aclanthology.org/2022.emnlp-demos.5",
pages = "38--49"
}

```

```

@inproceedings{loureiro-etal-2022-timelms,
  title = "{T}ime{LM}s: Diachronic Language Models from {T}witter",
  author = "Loureiro, Daniel and
    Barbieri, Francesco and
    Neves, Leonardo and
    Espinosa Anke, Luis and
    Camacho-collados, Jose",
  booktitle = "Proceedings of the 60th Annual Meeting of the Association fo
  month = may,
  year = "2022",
  address = "Dublin, Ireland",
  publisher = "Association for Computational Linguistics",
  url = "https://aclanthology.org/2022.acl-demo.25",
  doi = "10.18653/v1/2022.acl-demo.25",
  pages = "251--260"
}

```



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